

Jihwan Hong

csjihwanh@snu.ac.kr · Google Scholar · csjihwanh.com

Research Interests

- Video Understanding and Generation
- World Modeling
- 3D/4D Reconstruction and Generation

Education

Seoul National University <i>M.S. in Artificial Intelligence</i>	2025 – Present
Korea University <i>B.S. in Computer Science and Engineering</i>	2019 – 2025

Research Experience

Research Assistant, AIDAS Lab, Seoul National University, Korea <i>Advisor: Prof. Jaeyoung Do</i>	Aug. 2025 – Present
Research Intern, AIDAS Lab, Seoul National University, Korea <i>Advisor: Prof. Jaeyoung Do</i>	Jan. 2025 – Aug. 2025
Research Intern, MLV Lab, Korea University <i>Advisor: Prof. Hyunwoo J. Kim</i>	Jun. 2023 – Dec. 2024
Research Intern, AI Lab, Korea University <i>Advisor: Prof. Dongsuk Yook</i>	Oct. 2020 – Jan. 2021

Publications

When Predicting Nothing Beats SAM 3: Revisiting Evaluation in Video Object Segmentation <i>40th Annual Conference on Neural Information Processing Systems (NeurIPS), Evaluations and Datasets Track</i> Jihwan Hong , Woohyeon Park, Jaeik Kim, Jaeyoung Do [†] Project · arXiv	2026
VIRST: Video-Instructed Reasoning Assistant for SpatioTemporal Segmentation <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i> Jihwan Hong , Jaeyoung Do [†] Project · Paper · arXiv · Code	2026
Report of the 5th PVUW Challenge: Towards More Diverse Modalities in Pixel-Level Understanding <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)</i> Chang Liu <i>et al.</i> Paper · arXiv	2026

Preprints & Manuscripts

Dynin-Omni: Omnimodal Unified Large Diffusion Language Model <i>Under review</i> Jaeik Kim, Woojin Kim, Jihwan Hong , Yejoon Lee, Sieun Hyeon, Mintaek Lim, Yunseok Han, Dogeun Kim, Hoeun Lee, Hyungeun Kim, Jaeyoung Do [†] Demo · Code · Model · arXiv	2026
DM3SR: Benchmarking Deep Multi-View 3D Spatial Reasoning <i>Under review</i> Jusang Oh, Jihwan Hong , Jaeyoung Do [†]	2026
SoccerNet 2024 Challenges Results <i>arXiv preprint</i> Anthony Cioppa <i>et al.</i> arXiv	2024

Awards & Honors

3rd Place, 5th PVUW Challenge MeViS-Audio Track <i>CVPR 2026 Pixel-level Video Understanding in-the-Wild (PVUW) Workshop</i> 3rd Place of MeViS-Audio Track of the 5th PVUW: VIRST-Audio Jihwan Hong , Jaeyoung Do [†] arXiv	2026
2nd Place, CVPRW 2024 SoccerNet Challenge: Multi-View Foul Recognition <i>CVPR 2024 CVsports Workshop</i> VF-VARS: Leveraging Video Foundation Models for Video Assistant Referee Systems Jihwan Hong [*] , Youngseo Kim [*] , Junseok Lee, Jimin Lee, Sehyung Kim, Hyunwoo J. Kim [†] Technical Report	2024
1st Place, The 14th KSME Student Creative Design Competition <i>The Korean Society of Mechanical Engineers; Minister of Science and ICT Award, Government of South Korea</i> Development of an AI-Based Automatic Seat Belt Adjustment Device Code	2023

Academic Service

Reviewer <i>International Conference on Learning Representations (ICLR)</i>	2027
Reviewer <i>IEEE Transactions on Multimedia</i>	2026

Selected Projects

Anatomy- and Action-Conditioned Colonoscopic World Model <i>SNU Boramae Medical Center & NVIDIA HK</i> Developing a colonoscopic I2V world model conditioned on an anatomical topology map and actions.	Jul. 2026 – Present
Technical Drawing Registration Automation <i>POSCO DX</i> Developed a vision-language-model-based workflow for autonomous technical drawing registration.	Apr. 2025 – Jun. 2025